

# Final project manuscript

Project: Reproducible AI Research Study

## 1. Prepare the purpose, prerequisites and materials

### Teach the learning cycle

Primary target: Explain the mechanism and terminology of final project manuscript

#### Session · Sequence and minutes · Resources

**Session:** Session 1 · Understand · 45 min

**Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

**Resources:** Sections 1–6; record the brief recall in the workbook.

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**Session:** Session 2 · Practise · 45 min

**Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

**Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

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**Session:** Session 3 · Investigate and transfer · 45 min

**Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

**Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

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This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

## Retrieval answer guidance

### Independent reproduction check

Without opening the old lesson, explain one key idea from Independent reproduction check. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of independent reproduction check

Reference example, not the only acceptable wording: Teacher explanation: Independent reproduction checks whether another operator can use the recorded artifacts to recover the stated computation without hidden help. Provide manifest, fixture data, code, environment and expected metric definition. Record assistance and mismatches; reproduction is not new-data replication.

### Writing results and limitations

Without opening the old lesson, explain one key idea from Writing results and limitations. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of writing results and limitations

Reference example, not the only acceptable wording: Teacher explanation: Results and limitations should state observed estimates before interpreting them. Separate measured comparisons from causal or deployment claims. Give counts, denominators, effect sizes, valid uncertainty and failed conditions. State missing evidence and preserve the original primary outcome.

### Negative results and claim revision

Without opening the old lesson, explain one key idea from Negative results and claim revision. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of negative results and claim revision

Reference example, not the only acceptable wording: Teacher explanation: Negative results report that the evidence did not meet a declared expectation. They can reveal design limits or useful constraints without proving no effect anywhere. Compare results with the original criterion and uncertainty assumptions. Preserve the failed hypothesis, discuss plausible mechanisms and label revised questions as new work.

Two 45-minute sessions. A final manuscript integrates question, methods, results, limitations and reproducibility. Each table and claim should point to the correct run. Use the supplied constructed scenarios to make the mechanism inspectable. Collect W4 before sharing solutions. Students should name intermediate evidence and limitations; classroom pacing and learning effectiveness remain unvalidated.

### Outcomes

Explain the mechanism and terminology of final project manuscript

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

## Preparation

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

## Prerequisite answer

A result depends on the inputs, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from predictions or unmeasured conditions. If the prior concept cannot be explained, use a short retrieval activity before this chapter.

## End goal for the pictorial case

Explain final project manuscript using the records and boundaries of this fictional investigation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

## 2. Explain and model the mechanism

A final manuscript integrates question, methods, results, limitations and reproducibility. Each table and claim should point to the correct run.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Audit consistency across abstract, figures, raw results and artifact manifest. Remove unsupported universal claims; label exploratory analyses and unassessed outcomes.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

### Term · Meaning

**Term:** Manuscript audit

**Meaning:** A consistency check across research records and prose

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**Term:** Run linkage

**Meaning:** Connecting reported numbers to exact results

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**Term:** Evidence alignment

**Meaning:** Agreement between claims and their supporting artifacts

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## Final project manuscript — evidence trace

### Input and scope · Check or calculation · Conclusion and limit

**Input and scope:** Abstract says 100% success; results table shows 18/20, and two failures are omitted from the figure.

**Check or calculation:** Audit consistency across abstract, figures, raw results and artifact manifest. Remove unsupported universal claims; label exploratory analyses and unassessed outcomes.

**Conclusion and limit:** Correct to 90% over 20 with two failures; restore figure and align abstract/table/run. Do not hide failures to preserve the headline.

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The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Correct to 90% over 20 with two failures; restore figure and align abstract/table/run. Do not hide failures to preserve the headline. Qualification: A manuscript is ready only when its claims and artifacts align; formatting alone cannot certify evidence. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the mechanism; reject a claim resting on unmeasured evidence.

## Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Abstract says 100% success; results table shows 18/20, and two failures are omitted from the figure.

Correct to 90% over 20 with two failures; restore figure and align abstract/table/run. Do not hide failures to preserve the headline.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

### Thinking move · Teacher prompt

**Thinking move:** Plan

**Teacher prompt:** What is given? What is the question? Which procedure fits these facts?

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**Thinking move:** Do

**Teacher prompt:** Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

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**Thinking move:** Check

**Teacher prompt:** Does the output answer the question? Which assumption or limitation prevents a larger claim?

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Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

### A pictorial worked case: Final project manuscript

Build a manuscript from claim to record. A capstone manuscript compares two frozen pattern classifiers on ten reserved cases.

#### The facts we will use

A is correct on 7/10; B on 8/10. All cases are indoor, from five plants with two views each.

AI ACADEMY / AI-7/34

### Indoor capstone photo outcomes

correct photos out of 10; five plants



Bar length shows the supplied value; the marked baseline is zero.

## Final project manuscript

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. State bounded claim

B adds one correct photo on this indoor set.



#### 2. Describe dependence

Ten photos come from five plants; report the grouping.



#### 3. Present results

Show 70% and 80% with counts and case-level outcomes.



#### 4. Preserve limitations

Do not claim outdoor robustness or ten independent plants.

### CHANGE ONE THING

The abstract calls the study a test on ten independent plants.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. State bounded claim

**What happens and why:** B adds one correct photo on this indoor set.

**Step:** 2. Describe dependence

**What happens and why:** Ten photos come from five plants; report the grouping.

### Step: 3. Present results

**What happens and why:** Show 70% and 80% with counts and case-level outcomes.

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### Step: 4. Preserve limitations

**What happens and why:** Do not claim outdoor robustness or ten independent plants.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

## Why this result follows

A final manuscript aligns question, procedure, evidence and scope. Every figure and conclusion should point back to the records that support it.

## What this example does not establish

No population superiority or significance claim is supplied; dependence and sample coverage need explicit treatment.

## Change one thing in the pictured case

Change: The abstract calls the study a test on ten independent plants.

Prediction: Correct it to five plants with two views each.

Reason: The source-object count, not the photo count, defines the independent plants.

## Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

# 3. Demonstrate and guide practice

### Time or stage · Teacher action

**Time or stage:** Session 1: 0–7 minutes

**Teacher action:** Retrieve the prerequisite; identify the input, procedure and evidence source.

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**Time or stage:** 7–18 minutes

**Teacher action:** Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.

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**Time or stage:** 18–32 minutes

**Teacher action:** Trace the declared procedure and worked case. Complete W2, recording intermediate steps.

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**Time or stage:** 32–45 minutes

**Teacher action:** Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.

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**Time or stage:** Session 2: 0–8 minutes

**Teacher action:** Retrieve the worked reasoning and collect the guided response before discussing it.

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**Time or stage:** 8–20 minutes

**Teacher action:** Compare a proposed boundary variant with the original case; identify what remains fixed.

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**Time or stage:** 20–35 minutes

**Teacher action:** Collect W4 independently. Require the procedure trace and the bounded conclusion.

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**Time or stage:** 35–45 minutes

**Teacher action:** Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

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Use the worked example in Sections 3 and 6. Collect a prediction before revealing its result; ask the learner to point to each intermediate fact.

Provide an input/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace. Extension: compare two concrete variants of this mechanism while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

## Reduce help deliberately

Completion task: Abstract says 100% success; results table shows 18/20, and two failures are omitted from the figure. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Correct to 90% over 20 with two failures; restore figure and align abstract/table/run. Do not hide failures to preserve the headline.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

### Individual reasoning check

Use the worked case for Final project manuscript. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A manuscript is ready only when its claims and artifacts align; formatting alone cannot certify evidence.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

### Pictorial practice question

Connect the manuscript’s 7/10 and 8/10 scores to the five-plant grouping. Which abstract claim must be corrected if it says ten independent plants?

### Reasoned teacher answer

B adds one correct photo on this indoor set. Ten photos come from five plants; report the grouping. Show 70% and 80% with counts and case-level outcomes. Do not claim outdoor robustness or ten independent plants. Changed case: Correct it to five plants with two views each. The source-object count, not the photo count, defines the independent plants.

Changed-case reasoning: Correct it to five plants with two views each. The source-object count, not the photo count, defines the independent plants.

Collect each learner’s explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

## 4. Run the investigation and diagnose misconceptions

The worked result establishes every future case: A manuscript is ready only when its claims and artifacts align; formatting alone cannot certify evidence.

A missing input may be replaced without recording the change: A replacement changes the evidence or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent evidence: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the evidence ledger. Add a boundary or missing-input case and record the expected

response before checking. More successful repeats of one case do not establish wider coverage.

## Match the support to the first error

### What the response shows · Next teaching action

**What the response shows:** Missing or misread facts

**Next teaching action:** Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

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**What the response shows:** Wrong procedure

**Next teaching action:** Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

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**What the response shows:** Arithmetic or implementation error

**Next teaching action:** Trace intermediate values or lines; preserve the first mismatch and retry that step.

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**What the response shows:** Correct result without a reason

**Next teaching action:** Ask for a trace and a changed case before marking independent understanding.

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**What the response shows:** An unsupported wider claim

**Next teaching action:** Ask which supplied evidence supports the claim and which further observation would be needed.

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Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

## 5. Assess and explain the answers

### Guided and exit questions

Guided: Correct to 90% over 20 with two failures; restore figure and align abstract/table/run. Do not hide failures to preserve the headline. Qualification: A manuscript is ready only when its claims and artifacts align; formatting alone cannot certify evidence. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the mechanism; reject a claim resting on unmeasured evidence.

## W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

Teacher explanation: A final manuscript integrates question, methods, results, limitations and reproducibility. Each table and claim should point to the correct run. Audit consistency across abstract, figures, raw results and artifact manifest. Remove unsupported universal claims; label exploratory analyses and unassessed outcomes.

## W2 Trace the worked case

Abstract says 100% success; results table shows 18/20, and two failures are omitted from the figure. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Correct to 90% over 20 with two failures; restore figure and align abstract/table/run. Do not hide failures to preserve the headline.

## W3 Audit the limit

A report omits this qualification: A manuscript is ready only when its claims and artifacts align; formatting alone cannot certify evidence. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A manuscript is ready only when its claims and artifacts align; formatting alone cannot certify evidence. State the observed result separately from the unverified or out-of-scope claim; preserve the original evidence rather than inventing a missing fact.

## W4 Apply a new case independently

Manuscript says 12 evaluated cases, raw record has 10 completed and 2 missing with 8 passes. Correct the summary.

Teacher explanation:  $8/10=80\%$  completed pass, coverage  $10/12$  and 2 unassessed; state all denominators. Do not imply 12 observed outcomes.

## W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked input or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure. Require preserved settings, original outputs and data roles. A manuscript is ready only when its claims and artifacts align; formatting alone cannot certify evidence. The new case should test this particular mechanism rather than merely restate the original result.

## Independent assessment and fresh retry

W4:  $8/10=80\%$  completed pass, coverage  $10/12$  and  $2$  unassessed; state all denominators. Do not imply  $12$  observed outcomes. Require intermediate reasoning and the stated scope. Fresh retry: Figure labels old corpus version while methods uses new one. Expected reasoning: Trace which run generated it, correct the version linkage and rebuild mismatched figures if needed. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

## Assess independent transfer

Selected case: Manuscript says  $12$  evaluated cases, raw record has  $10$  completed and  $2$  missing with  $8$  passes. Correct the summary.

## Selected case answer

Teacher explanation:  $8/10=80\%$  completed pass, coverage  $10/12$  and  $2$  unassessed; state all denominators. Do not imply  $12$  observed outcomes.

### Dimension · Evidence required

**Dimension:** Explanation

**Evidence required:** Names the applicable idea or procedure and ties it to the supplied facts.

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**Dimension:** Trace and result

**Evidence required:** Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

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**Dimension:** Limits

**Evidence required:** States the assumptions, missing evidence or conditions under which the result would change.

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Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

## Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

# 6. Connect the project, support and next lesson

For Reproducible AI Research Study, use this week's independent task as a rehearsal for the final evidence pack: Manuscript says 12 evaluated cases, raw record has 10 completed and 2 missing with 8 passes. Correct the summary. Keep actual observations separate from expected results, preserve failures, compare a baseline and explain the limits.

Provide an input/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace. Extension: compare two concrete variants of this mechanism while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Presentation and challenge session. Preserve the current case record and retrieve its evidence boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

## Plan the delayed return

Suggested return points in this level: ai-7/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

### Record · Teacher evidence

**Record:** Learner code and date

**Teacher evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Record:** Concept and case used; prior exposure

**Teacher evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Record:** Explanation / trace / limits

**Teacher evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Record:** Support and first repair

**Teacher evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Record:** Changed case and delayed outcome

**Teacher evidence:** \_\_\_\_\_

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**Record:** Next teaching action

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### **Project evidence from this case**

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.